

Project Part 2: Documentation Document

1. Brief explanation of your approach for complex queries (queries 5, 6, 7)

Query 5: Book Popularity Report

Approach:

This query analyzes how popular each book is by counting how many times it has been borrowed. The books table is joined with the loan table using an **INNER JOIN** to ensure only books that were loaned are included. A **LEFT JOIN** is used with the review table so that books are still shown even if they do not have any reviews. Aggregate functions are applied to count the number of loans and calculate the average rating. The **GROUP BY** clause groups results by book details, while the **HAVING** clause filters the output to include only books that were loaned at least twice.

Query 6: Member Reading History

Approach:

This query retrieves the complete borrowing history for each member, including currently borrowed and previously returned books. A **LEFT JOIN** is used between members and loans so that members appear even if they have no loans. Additional **LEFT JOINs** with books and review ensure that book details and review information are included when available. Reviews are matched using both `member_id` and `book_id` to ensure accuracy. The results are ordered by member name and loan date to present a clear chronological reading history.

Query 7: Revenue Analysis by Genre

Approach:

This query evaluates financial performance by grouping loan and payment data by book genre. An **INNER JOIN** between books and loan ensures only borrowed books are considered. A **LEFT JOIN** with the payment table allows loans without payments to still appear in the results. Aggregate functions are used to calculate the total number of loans, total fines collected, and average fine per loan for each genre. The **ISNULL** function is applied to handle cases where no payments exist, ensuring accurate numerical results.

2. Screenshots of query results for at least 5 different queries

-- 3. Overdue Loans with Member Details

	MembersNames	phone_number	book_title	LibraryName	days_overdue	total_fines_paid
1	Ali Al-Saadi	92558747	Modern Science	Muscat central library	44	4.00
2	Fatima Al-Arthy	78778741	Physics 101	Sohar city library	35	5.40
3	Layla Al-Zaabi	90014158	The Lost City	Sohar city library	33	4.60
4	Mohammed Al-Hosni	99661101	The Great Adventure	Muscat central library	46	5.00
5	Noor Al-Harthy	93334455	History of Oman	Sohar city library	38	4.25
6	Salim Al-Hinai	91234567	Business Basics	Sohar city library	41	5.75

--8. Monthly Loan Statistics

	MonthName	Total_Loans	Total_Returned	Total_Issued_Overdue
1	November	15	5	10
2	December	1	1	0

--11. High-Value Books Analysis

	b_title	genre	price	Genre_Avg_Price	Price_Above_Avg
1	Junior Encyclopedia	Children	19.95	15.750000	4.200000
2	The Lost City	Fiction	27.00	21.783333	5.216667
3	Fantasy Land	Fiction	25.00	21.783333	3.216667
4	Mystery House	Fiction	22.00	21.783333	0.216667
5	Ancient Civilizations	Non-fiction	38.00	30.200000	7.800000
6	History of Oman	Non-fiction	35.00	30.200000	4.800000
7	Programming SQL	Reference	60.00	50.000000	10.000000
8	Data Science Intro	Reference	55.00	50.000000	5.000000

--13. vw_CurrentLoans

	loan_id	Member_Name	Book_Title	loan_date	due_date	statuss	Days_Until_Due
1	6	Mohammed Al-Hosni	The Great Adventure	2025-11-01	2025-11-15	Issued	-46
2	7	Fatima Al-Arthy	World History	2025-11-02	2025-11-16	Overdue	-45
3	9	Layla Al-Zaabi	Kids Stories	2025-11-04	2025-11-18	Issued	-43
4	10	Ahmed Al-Mansoori	Mystery House	2025-11-05	2025-11-19	Overdue	-42
5	12	Aisha Al-Rashdi	Math Guide	2025-11-07	2025-11-21	Issued	-40
6	13	Khalid Al-Balushi	Fairy Tales	2025-11-08	2025-11-22	Overdue	-39
7	15	Yousef Al-Kindi	Fantasy Land	2025-11-10	2025-11-24	Issued	-37
8	16	Mohammed Al-Hosni	Health & Life	2025-11-11	2025-11-25	Issued	-36
9	18	Ali Al-Saadi	Bedtime Stories	2025-11-13	2025-11-27	Overdue	-34
10	19	Layla Al-Zaabi	The Lost City	2025-11-14	2025-11-28	Issued	-33

--15. vw_BookDetailsWithReviews

	b_title	genre	availability_status	Avg_Rating	Total_Reviews	Latest_Review_Date
1	Ancient Civilizations	Non-fiction	true	NULL	0	NULL
2	Bedtime Stories	Children	true	NULL	0	NULL
3	Business Basics	Non-fiction	true	4	1	2025-11-25
4	Creative Writing	Fiction	true	NULL	0	NULL
5	Data Science Intro	Reference	false	NULL	0	NULL
6	Fairy Tales	Children	true	3	1	2025-11-27
7	Fantasy Land	Fiction	false	5	1	2025-11-29
8	Health & Life	Non-fiction	true	NULL	0	NULL
9	History of Oman	Non-fiction	true	4	1	2025-11-28
10	Junior Encyclopedia	Children	true	NULL	0	NULL
11	Kids Stories	Children	true	5	1	2025-11-23
12	Math Guide	Reference	false	5	1	2025-11-26
13	Modern Science	Reference	false	3	1	2025-11-22
14	Mystery House	Fiction	false	2	1	2025-11-24
15	Physics 101	Reference	false	NULL	0	NULL
16	Poetry Nights	Fiction	true	NULL	0	NULL
17	Programming SQL	Reference	true	NULL	0	NULL
18	The Great Advent...	Fiction	true	5	1	2025-11-20
19	The Lost City	Fiction	false	NULL	0	NULL
20	World History	Non-fiction	true	4	1	2025-11-21

3. Explanation of how your stored procedures handle edge cases

The stored procedures in the system are designed to ensure **data integrity, reliability, and correctness** by handling common edge cases that may occur during database operations.

1. Invalid or Missing Foreign Keys

Stored procedures validate input IDs (such as `member_id`, `book_id`, or `loan_id`) before performing inserts or updates. This prevents operations on non-existent members, books, or loans and avoids foreign key constraint violations.

2. Duplicate or Repeated Actions

Procedures include checks to prevent duplicate operations, such as:

- Issuing the same book multiple times while it is already on loan.
- Recording multiple returns for the same loan.

These checks ensure logical consistency in loan and book status.

3. NULL and Optional Data Handling

Edge cases where optional data is missing (for example, no payment made or no review submitted) are handled using default values or conditional logic. This allows procedures to continue executing without errors while keeping data accurate.

4. Date and Status Validation

Stored procedures validate date-related edge cases by ensuring:

- Return dates are not earlier than loan dates.
- Due dates are valid.
- Loan status updates (Issued, Returned, Overdue) are applied consistently based on business rules.

5. Financial Edge Cases

When handling payments, procedures ensure:

- Payment amounts are greater than zero.
- Payments are only linked to valid loans.
- Loans without fines do not incorrectly generate revenue records.

6. Transaction Safety

Critical procedures use transactions so that if any step fails, all changes are rolled back. This prevents partial updates and keeps the database in a consistent state.

7. Graceful Handling of Empty Results

Procedures are written to return meaningful results even when no matching records exist (e.g., no overdue loans or no payments for a genre), avoiding runtime errors and ensuring stable reporting.

Summary

By validating inputs, enforcing business rules, handling NULL values, and using transactions, the stored procedures safely manage edge cases and maintain data accuracy across the library management system.